

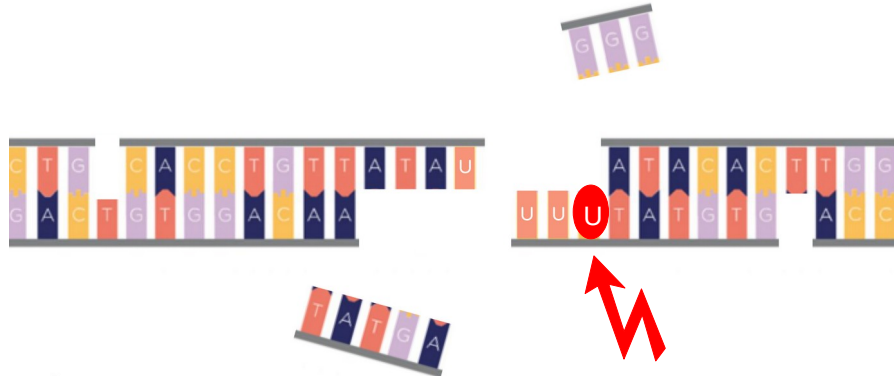
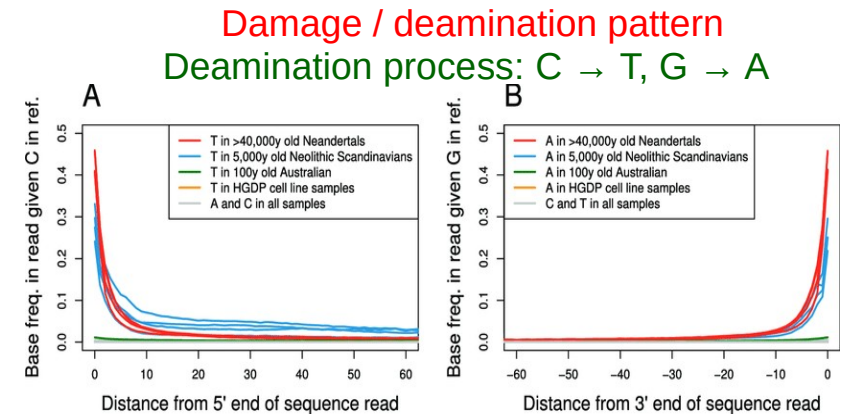
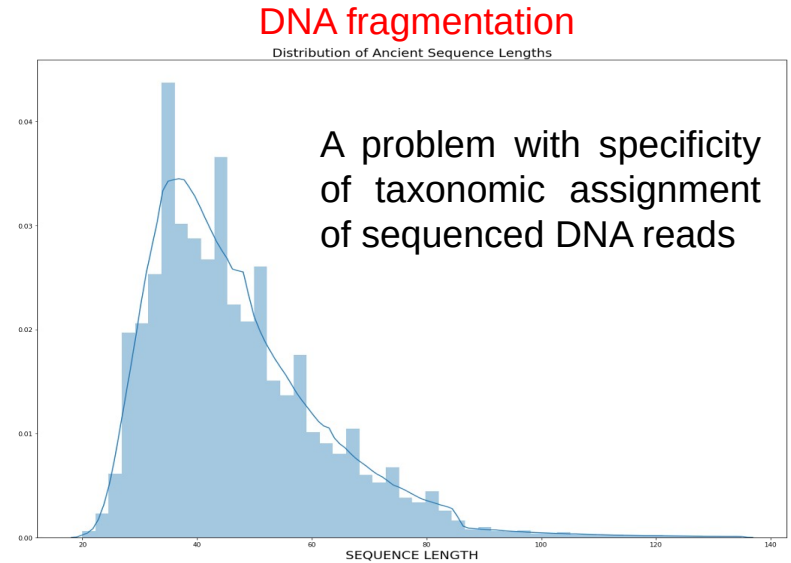


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CAGGAGGGCAGCGCCCTGGGCAACCCTATGTAGATGAAGCTGCCGAGAGGATCAAAGAACCAGACAGGAGGAAAGAGGCGGTGAAG
TCTCCTGTCTCATCCCTTAGGAAGCCTGAGGAGATGGGTAAGGGCATTAGAAAGCCTCGAACCCAGGGGCAGAGGATAGTTGTAGGCA

Peculiarity: very few samples with little amounts of DNA, hence hard for statistical analysis

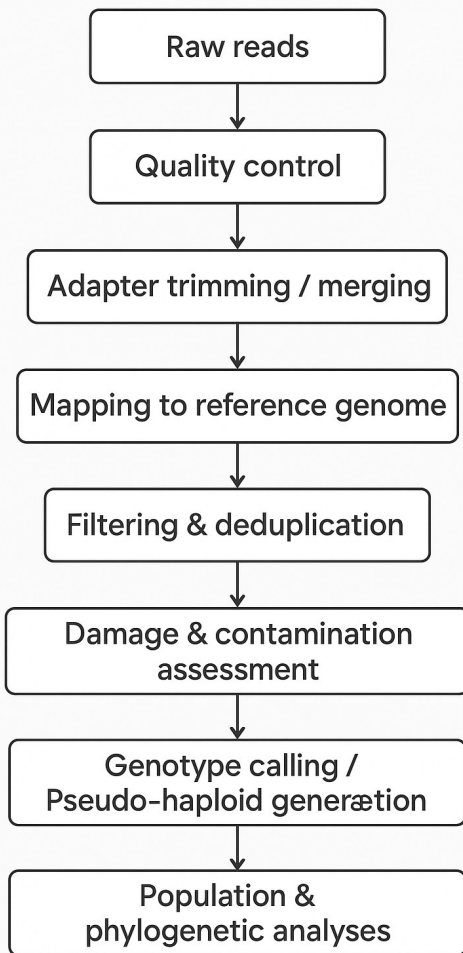
Low statistical evidence to make confident conclusions, predictions can rarely be validated

Ancient DNA (aDNA) degradation





Ancient DNA (aDNA) Analysis Workflow



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Ancient DNA analysis

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Abstract

Although the first ancient DNA molecules were extracted more than three decades ago, the first ancient nuclear genomes could only be characterized after high-throughput sequencing was invented. Genome-scale data have now been gathered from thousands of ancient archaeological specimens, and the number of ancient biological tissues amenable to genome sequencing is growing steadily. Ancient DNA fragments are typically ultrashort molecules and carry extensive amounts of chemical damage accumulated after death. Their extraction, manipulation and authentication require specific experimental wet-laboratory and dry-laboratory procedures before patterns of genetic variation from past individuals, populations and species can be interpreted. Ancient DNA data help to address an entire array of questions in anthropology, evolutionary biology and the environmental and archaeological sciences. The data have revealed a considerably more dynamic past than previously appreciated and have revolutionized our understanding of many major prehistoric and historic events. This Primer provides an overview of concepts and state-of-the-art methods underlying ancient DNA analysis and illustrates the diversity of resulting



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Ancient Metagenomic Studies: Considerations for the Wider Scientific Community

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ABSTRACT

Like modern metagenomics, ancient metagenomics is a highly data-rich discipline, with the added challenge that the DNA of interest is degraded and, depending on the sample type, in low abundance. This requires the application of specialized measures during molecular experiments and computational analyses. Furthermore, researchers often work with finite sample sizes, which impedes optimal experimental design and control of confounding factors, and with ethically sensitive samples necessitating the consideration of additional guidelines. In September 2020, early career researchers in the field of ancient metagenomics met (Standards, Precautions & Advances in Ancient Metagenomics 2 [SPAAM2] community meeting) to discuss the state of the field and how to



Annual Review of Genomics and Human Genetics

A Robust Framework for Microbial Archaeology

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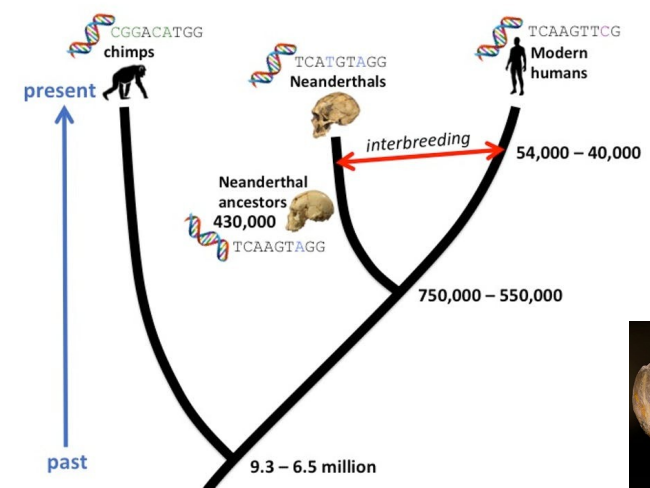
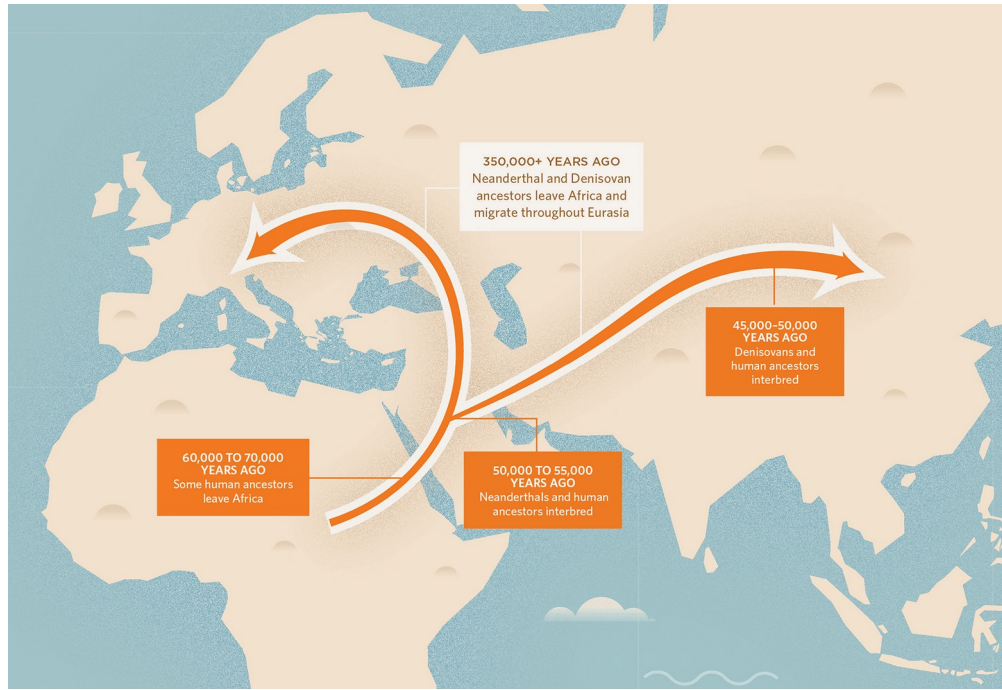
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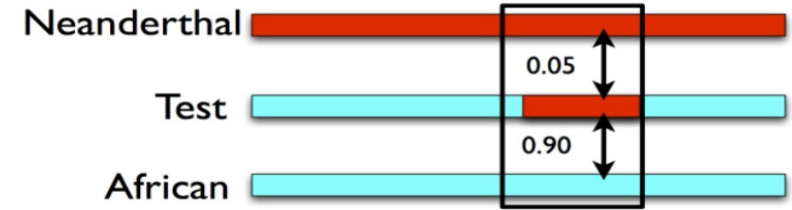
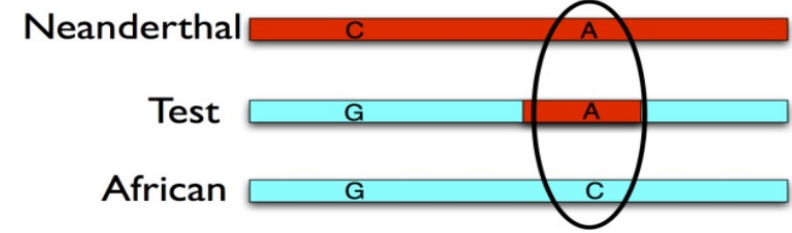
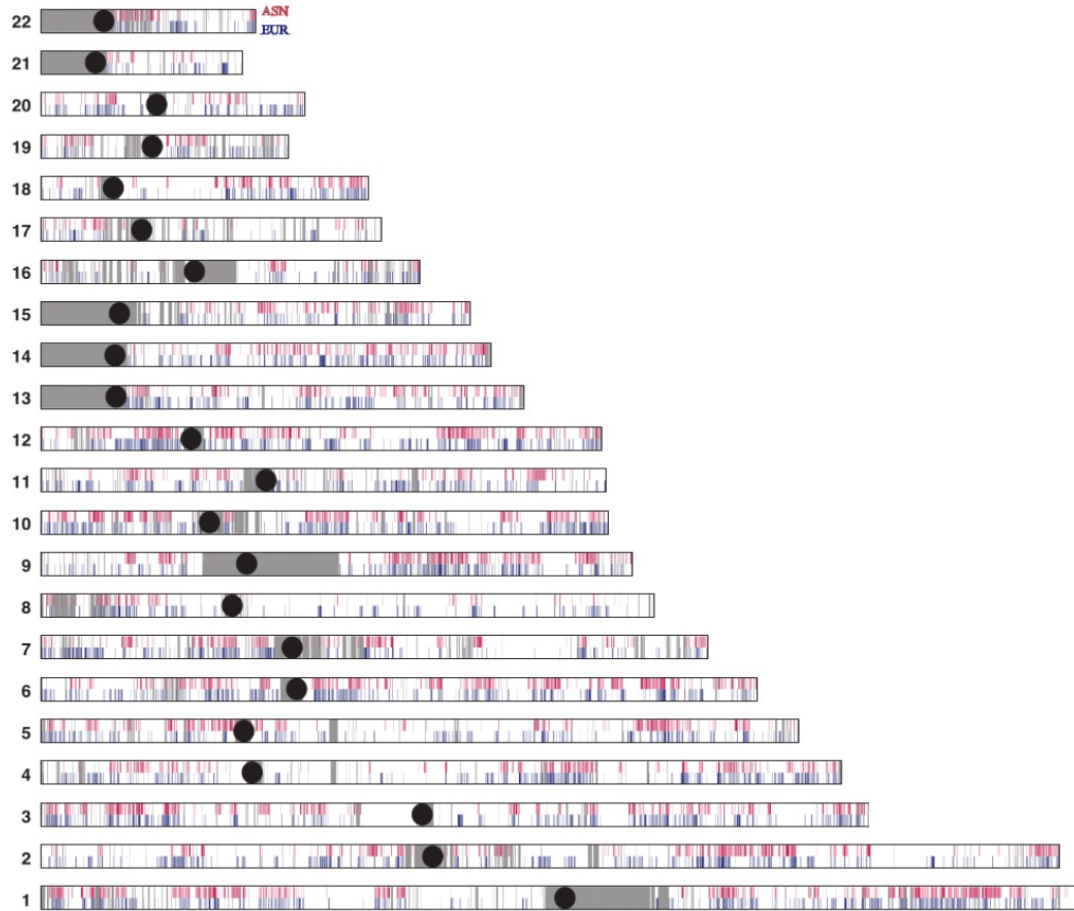


Neanderthal DNA in Modern Human Genomes Is Not Silent

From skin color to immunity,
human biology is linked to our archaic ancestry

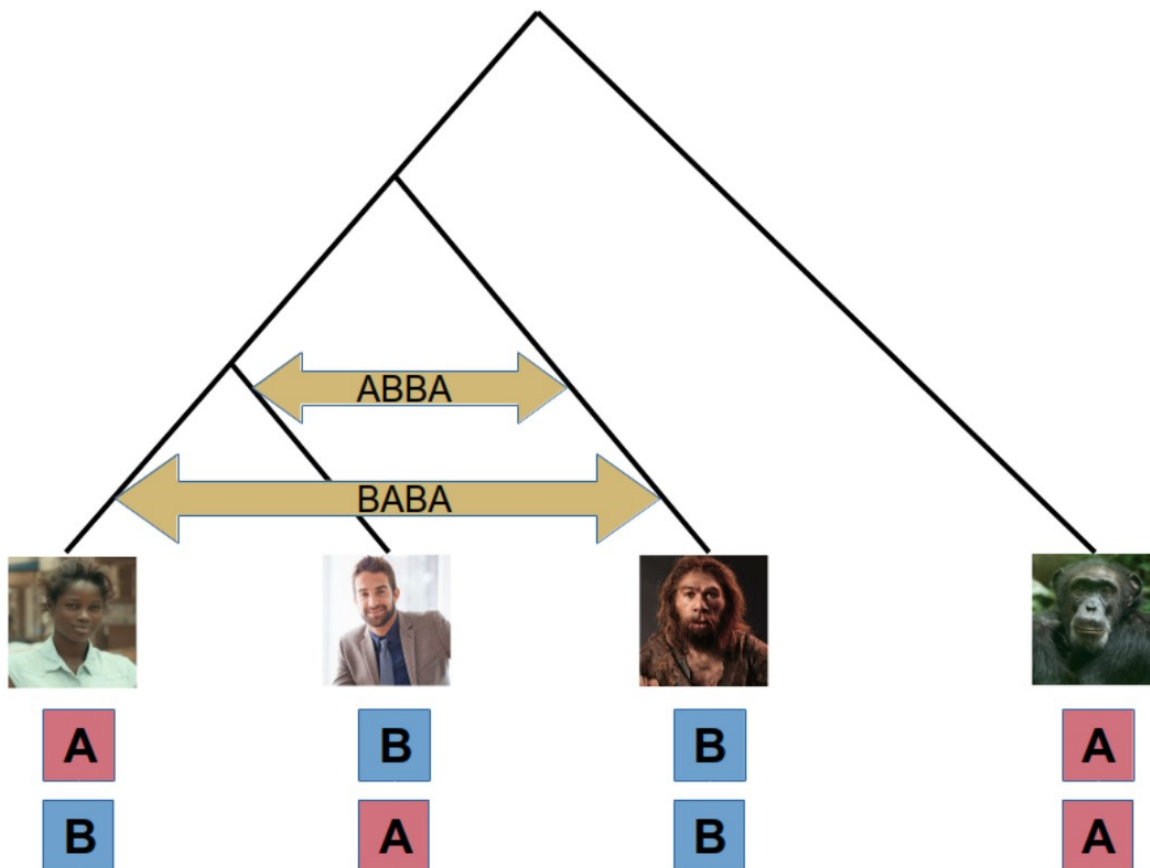


2010 Draft Neanderthal Genome Sequenced



Conditional Random Fields (CRF): David Reich et al., Nature 2014, 2016
 S* and S' statistics: Benjamin Vernot and Joshua Akey, Science 2014, 2016

F4-statistic, D-statistic: ABBA-BABA test



- Four populations considered: Ape is an outgroup, Europeans and Africans are tested for their gene flow with Neanderthals
- A – ancestral allele, B – derived allele
- Count number of sites corresponding to ABBA and BABA situations

$$D = \frac{N_{ABBA} - N_{BABA}}{N_{ABBA} + N_{BABA}}$$

- N_{ABBA} the total counts of ABBA patterns, N_{BABA} the total counts of BABA patterns
- Excess of shared derived alleles between Europeans and Neanderthals indicates a gene flow between them
- Positive values of D imply Neanderthals are closer to Europeans
- Negative values of D imply Neanderthals are closer to Africans

Example of D-statistic computation from 1000G project

We will use ANGSD (Korneliussen T, 2014) and compute D-statistic in order to test potential Neanderthal introgression to European (CEU) and African (YRI) populations from the 1000G project.

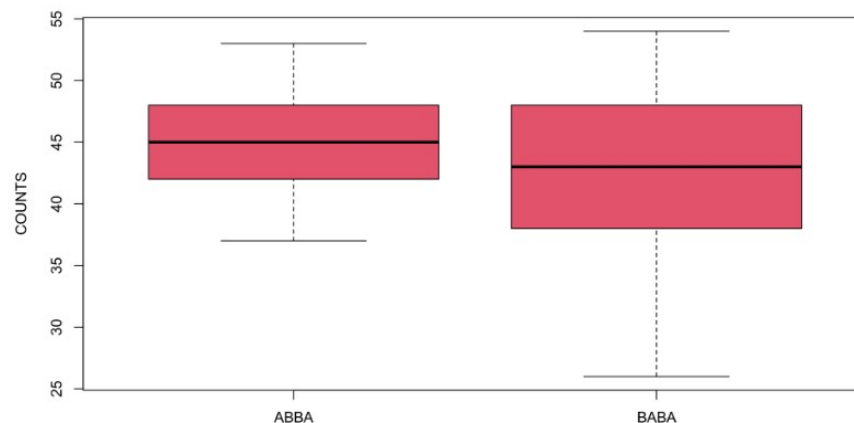
```
1 echo Neandertal.bam > bams
2 find $PWD -name '*YRI*.bam' | head -10 >> bams
3 find $PWD -name '*CEU*.bam' | head -10 >> bams
4 angsd -out out -doAbbababa 1 -bam bams -doCounts 1 -anc anc.fa
5 Rscript jackKnife.R file=out.abbababa indNames=bams outfile=out_abbababa_results
```

and, finally, we will display the resulting table of D-statistics and plot the ABBA and BABA counts using custom R scripts:

```
1 df <- read.delim(file.path(path, "out_abbababa_results.txt"), header = T, sep = "\t")
2 df <- df[grepl("Neandertal", as.character(df$H3)), ]
3 df <- df[grepl("CEU", as.character(df$H2)), ]
4 df <- df[grepl("YRI", as.character(df$H1)), ]
5 YRI <- matrix(unlist(strsplit(as.character(df$H1), "\\.")), ncol = 9, byrow = T)
6 df$H1 <- paste(YRI[, 6], YRI[, 2], sep = "_")
7 CEU <- matrix(unlist(strsplit(as.character(df$H2), "\\.")), ncol = 9, byrow = T)
8 df$H2 <- paste(CEU[, 6], CEU[, 2], sep = "_")
9 df$H3 <- "NEAND"
10 df[1:8, 1:8]
```

	H1	H2	H3	nABBA	nBABA	Dstat	jackEst	SE
11	YRI_NA19102	CEU_NA12342	NEAND	51	46	0.05154639	0.05154639	0.1220787
12	YRI_NA19213	CEU_NA12342	NEAND	38	31	0.10144930	0.10144930	0.1286760
13	YRI_NA18504	CEU_NA12342	NEAND	50	53	-0.02912621	-0.02912621	0.1353893
14	YRI_NA19108	CEU_NA12342	NEAND	38	42	-0.05000000	-0.05000000	0.1393287
15	YRI_NA19238	CEU_NA12342	NEAND	44	38	0.07317073	0.07317073	0.1305890
16	YRI_NA19102	CEU_NA12348	NEAND	46	45	0.01098901	0.01098901	0.1076116
17	YRI_NA19213	CEU_NA12348	NEAND	37	34	0.04225352	0.04225352	0.1122396
18	YRI_NA18504	CEU_NA12348	NEAND	42	49	-0.07692308	-0.07692308	0.1068372

```
1 boxplot(df$nABBA, df$nBABA, names = c("ABBA", "BABA"), ylab = "COUNTS", col = 2)
```



```
1 wilcox.test(df$nABBA, df$nBABA)
```

Wilcoxon rank sum test with continuity correction

data: df\$nABBA and df\$nBABA

W = 367, p-value = 0.2933

alternative hypothesis: true location shift is not equal to 0

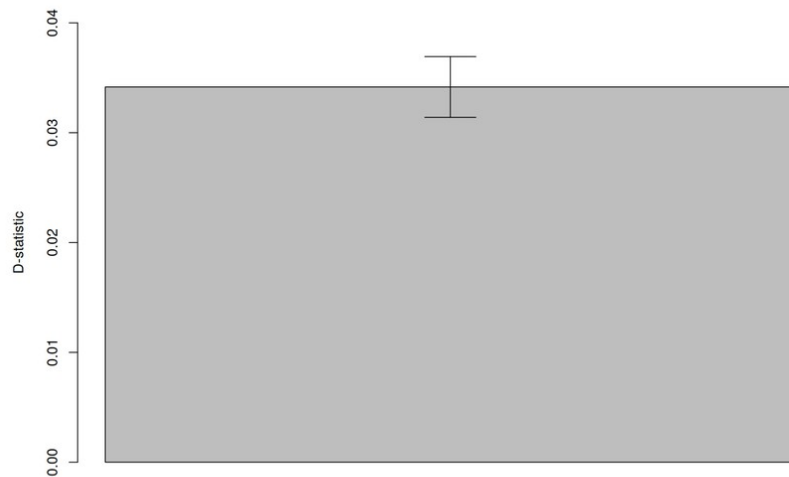
D-statistic computation for higher coverage samples

D-statistic analysis requires large number of genomic sites, therefore the low-coverage 1000G data are not optimal. Now, we are going to use French and Yoruba samples from the **Draft Neanderthal genome** project where Neandertal introgression was first detected:

	H1	H2	H3	nABBA	nBABA	Dstat	jackEst
1	Yoruba.bam	French.bam	Neandertal.bam	267387	249721	0.03416308	0.03416308
2	Neandertal.bam	French.bam	Yoruba.bam	732198	249721	0.49136130	0.49136130
3	Neandertal.bam	Yoruba.bam	French.bam	732198	267387	0.46500400	0.46500400

	SE	Z
1	0.002764548	12.35756
2	0.002449853	200.56760
3	0.002868187	162.12470

First line of the output corresponds to the tree displayed on the previous slide, i.e. where we test for potential gene flow from Neanderthals to modern Europeans or Africans. Large number of genomic sites ensures confident positivity of the D-statistic.



Positive D-statistic with errors bars not overlapping zero implies a gene flow between Neanderthals and modern Europeans.

